



Mapping a Century of Urban Development and Management Research: Bibliometric Structure, Thematic Development, and Emerging Research Directions

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Received: 07-17-2026

Revised: 09-12-2026

Accepted: 09-19-2026

Citation: M. Saqlain and J. M. Merigo, "Mapping a century of urban development and management research: Bibliometric structure, thematic development, and emerging research directions," *J. Urban Dev. Manag.*, vol. 5, no. 3, pp. 240–266, 2026. <https://doi.org/10.56578/judm050305>.



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Abstract: Urban development and management (UDM) plays a central role in addressing the environmental, social, and governance challenges associated with rapid urbanization. Although bibliometric studies have examined specific areas of urban research, the broader intellectual structure and long-term thematic development of UDM remain fragmented across multiple research streams. This study investigates the evolution, knowledge structure, and emerging research directions of UDM through a large-scale bibliometric and science-mapping analysis. A dataset of 40,435 journal articles and reviews published between 1925 and 2025 was retrieved from Scopus and analyzed following the Scientific Procedures and Rationales for Systematic Literature Reviews (SPAR-4-SLR) protocol. Performance analysis, co-citation analysis, bibliographic coupling, keyword co-occurrence analysis, and thematic mapping were conducted using bibliometrix and VOSviewer. The results showed a marked acceleration in research output, with 45.9% of all documents published during 2021–2025. The dataset accumulated 1,285,702 citations, averaging 31.8 citations per document. China recorded the largest publication output (9,309 documents), whereas the United States received the highest total citations (348,371) and h-index (223). Journal co-citation patterns revealed a multidisciplinary knowledge base centered on environmental and water sciences, while country-level bibliographic coupling showed a core-periphery structure organized around China, the United States, and Europe. Sustainable development and urban planning emerged as broad foundational themes, whereas water management occupied a central and well-developed position. The science-mapping results further identified climate adaptation, blue-green infrastructure, nature-based solutions, and digitally enabled urban management as prominent recent research directions. These findings demonstrate how UDM has developed from a predominantly planning- and growth-oriented field into a multidisciplinary research domain increasingly concerned with environmental limits, climate resilience, and data-supported urban governance. The study provides an integrated knowledge map of UDM and identifies research priorities concerning the Global South, sustainability measurement, blue-green infrastructure, climate-health interactions, digital urban management, circular urban systems, and urban governance.

Keywords: Urban development and management; Bibliometric analysis; Science mapping; Urban sustainability; Urban planning; Research trends

1 Introduction

In 2018, 55% of the world's population lived in urban areas, and this proportion is projected to reach 68% by 2050, adding approximately 2.5 billion people to urban populations, predominantly in Asia and Africa [1]. Urban land expansion is occurring even faster than population growth. Seto et al. [2] projected that global urban land cover could nearly triple between 2000 and 2030, with substantial consequences for biodiversity, carbon storage, and water systems. Cities already concentrate much of the world's economic activity, energy consumption, and greenhouse gas emissions, placing them at the center of efforts to achieve global sustainability [3, 4]. This role is reflected in Sustainable Development Goal (SDG) 11, which calls for inclusive, safe, resilient, and sustainable cities [5], and in the New Urban Agenda, which provides a broader framework for planning, financing, and governing urban development [6].

Urban development and management (UDM) lies at the intersection of these challenges. It concerns not only the physical growth and transformation of cities but also the institutional, economic, environmental, and managerial conditions under which urban change occurs. Consequently, UDM research extends across a wide range of concepts, methods, and urban problems, producing a large and increasingly diverse body of literature. A Scopus search conducted for the present study initially returned more than 72,000 records; after screening, 40,435 peer-reviewed journal articles and reviews published between 1925 and 2025 were retained. At this scale, conventional narrative review methods alone are insufficient for tracing long-term changes in research output, identifying the works and research communities that have shaped the field, or examining how its thematic priorities have developed over time.

Bibliometric analysis provides a systematic means of addressing these questions by quantifying research performance and mapping the intellectual, social, and conceptual structure of a field using bibliographic and citation data [7, 8]. Co-citation analysis can reveal the intellectual foundations shared by a research community, bibliographic coupling can identify relationships among contemporary research streams, and keyword co-occurrence analysis can characterize the conceptual organization of a field. Thematic mapping and temporal interpretation of these structures can further distinguish established, specialized, and developing areas of research [9, 10]. Such analyses are increasingly conducted using tools including the bibliometrix R package and VOSviewer, which allow large bibliographic datasets to be examined systematically [11–13].

Bibliometric reviews have already provided detailed accounts of several important branches of urban research, including smart cities [14] and urban sustainability assessment [15]. These studies have substantially clarified the development of individual research streams, but the broader UDM literature remains dispersed across planning, environmental management, infrastructure, governance, land-use, water, and technology-oriented domains. A field-level perspective is therefore needed not because these subjects have been unexplored, but because their relationships within the wider UDM knowledge structure remain insufficiently integrated. Issues such as stormwater management, water supply, land-use conversion, infrastructure provision, and service delivery frequently cross conventional disciplinary boundaries. Examining them within a common analytical framework makes it possible to determine how the intellectual foundations of UDM are connected, how its thematic organization has changed over a century of research, and which directions have become increasingly prominent.

Against this background, this study examines the bibliometric structure, thematic development, and emerging research directions of UDM literature indexed in Scopus between 1925 and 2025. Six research questions (RQs) were formulated following the Scientific Procedures and Rationales for Systematic Literature Reviews (SPAR-4-SLR) protocol [16] and the bibliometric guidelines of Donthu et al. [8]. RQ1 examines the publication and citation structure of UDM research. RQ2 evaluates the overall performance and development of the field. RQ3 identifies its most highly cited articles and influential documents, while RQ4 examines the most productive authors, institutions, and countries. RQ5 investigates the intellectual and research-community structure of UDM through co-citation analysis and bibliographic coupling. Finally, RQ6 examines its keyword and thematic structure, with particular attention to changes in research emphasis and the identification of prominent recent directions.

The study makes three main contributions. First, it provides a century-scale assessment of UDM based on more than 40,000 documents, allowing the growth and changing composition of the field to be examined beyond the shorter time horizons commonly adopted in topic-specific reviews. Second, it brings together performance indicators and multiple science-mapping approaches to show how major areas such as sustainable development, urban planning, water management, environmental protection, land-use change, and digitally supported urban management are positioned within the wider UDM knowledge structure. Third, the analysis connects these structural patterns with changes in thematic emphasis and develops a research agenda around areas requiring further attention, particularly sustainability measurement, climate adaptation, blue–green infrastructure, digital urban management, urban governance, and the geographical imbalance between established research centers and rapidly urbanizing regions.

The remainder of the paper is organized as follows. Section 2 reviews the conceptual development of UDM and previous bibliometric studies of urban research. Section 3 describes the data collection and analytical methods. Section 4 presents the performance and metric analyses, while Section 5 examines the science-mapping results. Section 6 integrates the principal findings, discusses thematic developments, and proposes a research agenda. Section 7 concludes the study and discusses its implications and limitations.

2 Literature Review

Early urban research approached development largely as a problem of physical planning, with land use, infrastructure, and urban form occupying central positions. From the 1970s onward, greater attention was given to the institutional processes through which cities are governed and transformed. Harvey [17] distinguished between managerial and entrepreneurial approaches to urban governance: the former emphasizes the provision of local services, whereas the latter concerns competition among cities for investment and economic growth. Planning theory underwent a related shift from technically oriented master planning toward negotiation and interaction among multiple planning stakeholders [18]. Together, these developments broadened the conceptual boundaries of UDM beyond land

use and infrastructure to include governance, finance, participation, and the institutional organization of urban change.

Sustainable development subsequently became a major organizing principle of urban research. The Brundtland Report placed sustainability firmly on the international policy agenda, while Campbell [19] framed the planning challenge around the tensions among economic growth, environmental protection, and social equity. These tensions remain central to contemporary UDM, although their interpretation has increasingly incorporated resilience alongside sustainability. Urban resilience concerns the capacity of urban systems to withstand disturbances, adapt to changing conditions, and respond across different spatial and temporal scales. Meerow et al. [20], after examining 25 definitions of urban resilience, proposed a synthesis that emphasized precisely these multiple scales, timeframes, and adaptive capacities. Sustainability and resilience therefore represent related but distinct perspectives through which the long-term management of urban systems has been examined.

Natural-resource management and the environmental consequences of urbanization constitute another major strand of UDM research. From an urban-metabolism perspective, cities function as systems that consume energy, water, and materials while generating wastes and emissions. Kennedy et al. [21] showed that the metabolism of many cities had intensified as per-capita resource consumption increased, placing greater pressure on energy, water, and material flows. Water management has been especially prominent within this literature. Brown et al. [22] described a progression in urban water regimes from the water-supply city toward the water-sensitive city, in which water management becomes integrated with urban design, ecology, and liveability. In parallel, approaches to sustainable urban drainage have developed under related concepts such as water-sensitive urban design, sustainable urban drainage systems, and low-impact development [23]. These developments illustrate how infrastructure management has progressively become connected with broader environmental and planning objectives.

Urban green space and ecosystem research extends this environmental perspective by examining the multiple functions that urban nature performs, including recreation, stormwater regulation, and contributions to public health. These functions, however, are not socially neutral. Greening initiatives can raise property values and contribute to the displacement of lower-income residents, introducing an important equity dimension into environmental management, as demonstrated by Wolch et al. [24]. Waste management, pollution control, and environmental protection constitute related areas in which UDM intersects with applied environmental science and engineering. Taken together, these strands show that urban environmental management increasingly involves not only the control of environmental pressures but also questions of distribution, accessibility, and social consequences.

Advances in geospatial technologies have also changed how urban development is observed and analyzed. Remote sensing and geographic information systems allow urban expansion, undeveloped land, and land conversion to be examined across large spatial extents and long periods [25]. Global evidence indicates that urban land expansion has been particularly rapid in developing regions and has frequently exceeded population growth, contributing to declining urban densities and increasing land consumption [26]. Research on China has been especially prominent because of the scale and speed of urbanization since the 1980s and the associated challenges involving land, housing, and environmental management [27]. More recently, digital technologies have introduced additional modes of inquiry into smart and data-driven urbanism, in which real-time information is used to observe, analyze, and manage urban systems [28]. This development links the established spatial-analysis tradition of UDM with newer forms of digitally supported urban management.

As the urban literature has expanded, bibliometric approaches have increasingly been used to examine its structure and development. Kamalski and Kirby [29] showed that keyword analysis could distinguish three broad areas of urban knowledge distributed across social-science, applied-science, and specialist urban journals. Subsequent studies increasingly focused on sustainability. Wang et al. [30] mapped sustainable-city research from 1992 to 2016 across the natural and social sciences, while Sharifi [15] analyzed 3,877 publications on urban sustainability assessment over approximately three decades using VOSviewer and Science Mapping Analysis Tool (SciMAT). Marvuglia et al. [31] examined advances in urban sustainability assessment in the context of the energy transition. Other studies investigated how cities have been conceptualized and labelled: Schraven et al. [32] mapped 35 city labels used between 1990 and 2019, and Ligorio et al. [33] applied topic modelling to distinguish research on sustainable cities from research on cities for sustainable development. Smart-city scholarship has likewise been mapped through combinations of co-citation clustering and text analysis to identify major development paths [34] and contrasting perspectives on the organization of sustainable smart urban ecosystems [35]. Urban resilience forms another substantial stream, examined through scientometric analysis of its evolution [36], its relationship with the SDG [37], and a review of 958 documents on resilient infrastructure published between 2000 and 2022 [38]. Regional bibliometric perspectives remain less common, although Almulhim et al. [39] mapped 415 publications on urbanization and environmental sustainability in Gulf Cooperation Council countries between 1980 and 2021.

A further group of reviews concentrates on urban water, green infrastructure, and land-use systems. Li et al. [40] reviewed the mechanisms through which green infrastructure contributes to stormwater management and examined policies supporting its implementation, including China's sponge-city programme. Subsequent bibliometric studies addressed optimization and resilience in low-impact development during 2004–2020 [41], the development of urban

rainfall–runoff pollution-control technologies in China during 2000–2020 [42], and 3,988 publications on green stormwater infrastructure during 2000–2021, a literature that grew at an annual rate of 28.61% [43]. Related mapping studies examined 3,806 publications on biodiversity in urban green spaces and identified 2009 as a turning point in publication growth [44], while the relationship between land-use change and food security was mapped using CiteSpace and VOSviewer [45]. Collectively, these studies provide detailed views of important UDM components, but their thematic specialization also illustrates how the broader literature has developed through partially separated research communities.

The scope of bibliometric research on cities expanded further in 2026, with recent reviews converging on sustainability, governance, and technology as important areas of change. At a broad level, Nakajima and Asatani [46] examined urban sustainability as an integrated research field, while other studies addressed urban informality [47], land-use sustainability indicators [48], and housing sustainability [49, 50]. Governance-oriented research has also shifted from conventional public consultation toward partnership-based forms of engagement [51], alongside growing attention to digital transformation and smart cities [52]. Technology-focused reviews have examined artificial intelligence in urban planning [53], the roles of the Internet of Things (IoT) and blockchain in urban infrastructure [54], and modular construction in smart-city development [55]. At the same time, urban renewal and cultural resilience have attracted increasing attention [56]. This recent literature reinforces the growing convergence of environmental sustainability, institutional governance, and digital technology within contemporary urban research.

Despite this expanding body of bibliometric work, the literature remains predominantly organized around individual subfields or specific urban problems, as illustrated by studies of smart cities and urban sustainability assessment [14, 15]. Many existing reviews also cover substantially shorter periods than the century-long development considered here, and database and analytical choices vary across studies. The present study therefore complements rather than duplicates this literature. It examines 40,435 Scopus-indexed documents published between 1925 and 2025 and combines performance analysis with journal co-citation analysis, country bibliographic coupling, keyword co-occurrence analysis, and thematic mapping. This integrated design makes it possible to examine UDM as a connected research domain, relate its major intellectual and conceptual structures to their long-term development, and identify research directions that have become increasingly prominent in the most recent period.

3 Methods

Bibliographic records were retrieved from the Scopus database on 30 June 2026. The search was conducted across the title, abstract, and keyword fields using “urban development” and “management” as the search terms. A deliberately broad search strategy was adopted because UDM is not confined to a single disciplinary category and relevant research extends across urban planning, environmental management, infrastructure, governance, land use, water management, and technology-oriented domains. The search was therefore designed to capture the wider multidisciplinary knowledge environment associated with UDM rather than restrict the corpus to a predefined set of urban subfields.

The initial search returned 72,413 records. Publications assigned to 2026 were excluded to ensure that all annual observations represented complete publication years, reducing the dataset to 65,791 records covering 1925–2025. The document type was then restricted to articles and reviews, leaving 46,063 records. Limiting the dataset to English-language publications reduced this number to 41,291, and restricting the source type to journals resulted in 40,581 records. Finally, only documents classified by Scopus as being in the final publication stage were retained because bibliographic information for articles in press may still change. After these filters were applied, the final analytical dataset contained 40,435 journal articles and reviews. Complete bibliographic information and cited references were exported in CSV format and analyzed using the bibliometrix R package through Biblioshiny and VOSviewer.

The methodological design followed the SPAR-4-SLR protocol proposed by Paul et al. [16], together with the bibliometric-analysis guidelines of Donthu et al. [8]. The procedure comprised the three SPAR-4-SLR stages of assembling, arranging, and assessing, with their corresponding sub-stages summarized in Figure 1.

During the assembling stage, UDM was defined as the review domain, and Scopus was selected as the bibliographic source. The acquisition stage covered publications from 1925 through 2025, with the database searched on 30 June 2026. The initial search retrieved 72,413 records, and successive restrictions based on publication year, document type, language, source type, and publication stage reduced the final analytical sample to 40,435 documents. No post-retrieval exclusion was applied on the basis of subjective judgments of individual-document relevance. This decision preserved a rule-based dataset and avoided selectively removing interdisciplinary publications that may connect UDM with adjacent environmental, technological, infrastructure, or resource-management research. The resulting corpus is therefore treated as an operational representation of the multidisciplinary UDM research landscape rather than as an exhaustive or exclusive definition of the field.

The six RQs guided the assessment stage and were operationalized through complementary performance-analysis and science-mapping techniques. RQ1–RQ4 were addressed through publication, citation, author, institutional,

and country-level performance indicators, whereas RQ5 and RQ6 were examined through co-citation analysis, bibliographic coupling, keyword co-occurrence, and thematic analysis.

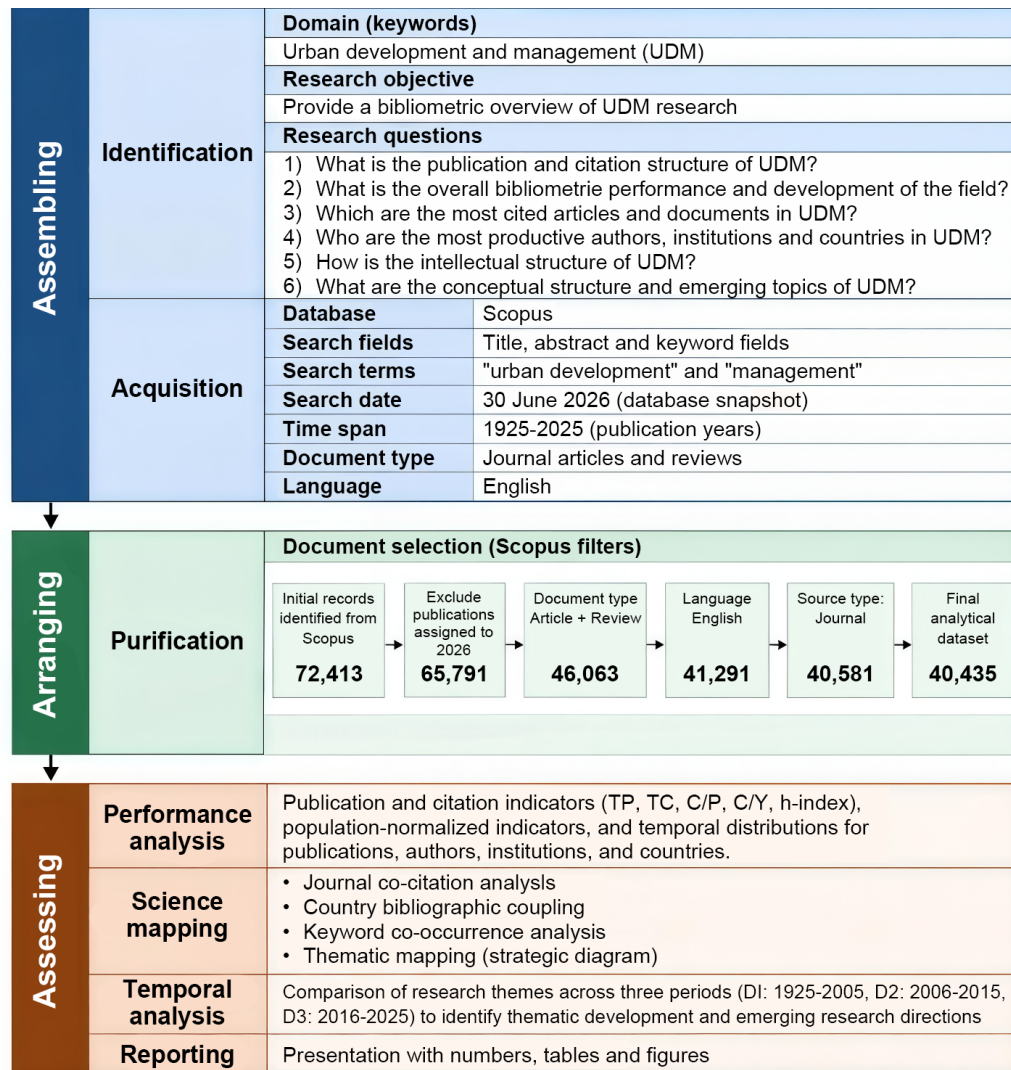


Figure 1. Procedure of the study based on the SPAR-4-SLR protocol

Note: SPAR-4-SLR = Scientific Procedures and Rationales for Systematic Literature Reviews; TP = total publications; TC = total citations; C/P = citations per publication; C/Y = citations per year; D1 = publications during 1925–2005; D2 = publications during 2006–2015; D3 = publications during 2016–2025.

The final dataset was assessed through two complementary analytical components. The first was performance analysis. Indicators included total publications (TP), total citations (TC), citations per publication (C/P), h-index, citations per year (C/Y), publication and citation intensity relative to population, and publication output across temporal periods. These indicators were used to characterize the development of the field and compare the contributions of publications, authors, institutions, and countries. Because raw citation counts are strongly affected by publication age, citations per year and temporal distributions were considered alongside cumulative citation indicators when interpreting research influence.

The second component was science mapping. Co-citation analysis was used to examine the intellectual foundations of UDM by identifying sources that were frequently cited together. Bibliographic coupling was used to examine similarities among contemporary research communities on the basis of shared references. Keyword co-occurrence analysis was applied to identify the conceptual structure of the literature and the relationships among major research topics. Thematic mapping was additionally used to distinguish broad, well-developed, specialized, and less-developed thematic configurations according to their relative centrality and density. These analyses were conducted using VOSviewer and bibliometrix, and the resulting structures were presented through network maps, thematic diagrams, tables, and descriptive statistics.

Changes in research emphasis were interpreted by combining the temporal distribution of publications and

keywords with the structures identified through keyword co-occurrence and thematic mapping. Accordingly, references to thematic development and emerging research directions in this study denote topics that became increasingly visible or prominent in the most recent publication period and occupied meaningful positions within the science-mapping results. They should not be interpreted as the output of a separate topic-modeling or citation-burst procedure.

Three methodological limitations should be recognized. First, bibliographic databases are dynamic, and citation counts, affiliations, indexing records, and document metadata may continue to change after the retrieval date. The present analysis therefore represents a snapshot of Scopus as accessed on 30 June 2026, while publication years were deliberately limited to 1925–2025 to avoid using an incomplete 2026 publication year. Second, the use of Scopus as the sole bibliographic database means that the results are influenced by its journal coverage, indexing policies, document classification, and metadata structure. Publications indexed exclusively in other databases, as well as excluded document types and non-English publications, are not represented. Third, citation-based indicators are sensitive to differences in publication age, discipline, document type, and citation practices. Older publications have had substantially more time to accumulate citations than recent studies; consequently, absolute citation counts were interpreted together with citations per year and temporal information rather than as direct measures of research quality.

4 Metric Analysis

Performance analysis provides the descriptive foundation of bibliometric research by quantifying patterns of scientific productivity and citation impact within a research field [8]. Based on the 40,435 documents retrieved from Scopus, the present analysis uses TP, TC, C/P, C/Y, and the h-index [57], together with temporal and population-normalized indicators where appropriate. These measures are used to address RQ1–RQ4 by evaluating publication and citation trends, document-level influence, and the productivity and impact of authors, institutions, and countries. The results provide a quantitative basis for the science-mapping analyses presented in Section 5.

Table 1. Annual numbers of papers published and citation patterns of UDM

Year	TP	TC	≥1000	≥500	≥100	≥50	≥25	≥10	≥5	≥1
1925–1950	2	1	0	0	0	0	0	0	0	1
1951–1975	81	1,032	0	0	3	4	5	11	20	44
1976–2000	2,827	53,552	1	8	110	259	505	979	1,344	2,108
2001	252	13,333	0	1	32	66	110	170	200	234
2002	226	13,543	1	6	32	54	94	137	171	207
2003	303	14,336	0	1	37	77	120	194	228	270
2004	371	16,365	0	2	43	91	172	248	284	333
2005	425	23,454	2	4	58	120	188	300	345	391
2006	477	21,893	0	3	54	118	209	305	367	437
2007	558	25,580	1	4	67	155	255	374	430	503
2008	601	31,493	3	4	73	153	261	398	477	562
2009	638	34,341	2	7	84	159	287	436	505	601
2010	809	45,471	2	10	118	233	391	558	658	767
2011	870	43,520	2	15	112	214	367	581	684	811
2012	893	41,527	1	5	99	218	371	589	714	840
2013	1,147	51,189	0	7	130	297	521	783	932	1,077
2014	1,237	57,767	4	9	122	283	507	841	1,010	1,162
2015	1,180	72,449	4	12	139	310	572	858	1,016	1,135
2016	1,266	69,211	2	15	150	349	640	956	1,091	1,222
2017	1,563	79,764	3	17	168	391	754	1,161	1,316	1,502
2018	1,703	74,504	0	7	172	408	790	1,272	1,486	1,654
2019	2,096	86,867	1	6	192	492	968	1,550	1,803	2,041
2020	2,360	92,842	0	7	204	556	1,075	1,714	2,015	2,299
2021	2,728	93,876	3	6	162	515	1,098	1,962	2,355	2,664
2022	2,877	72,710	0	2	110	345	851	1,757	2,311	2,770
2023	3,155	63,205	0	0	76	273	748	1,741	2,383	3,027
2024	4,228	58,568	0	0	42	213	611	1,789	2,808	3,947
2025	5,562	33,309	0	0	5	46	223	1,027	2,136	4,484
Total	40,435	1,285,702	32	158	2,594	6,399	12,693	22,691	29,089	37,093
%	100	–	0.08	0.39	6.42	15.83	31.39	56.12	71.94	91.73

Note: UDM = urban development and management; TP = total publications; TC = total citations; ≥1000, ≥500, ≥100, ≥50, ≥25, ≥10, ≥5, and ≥1 indicate the numbers of publications receiving at least the corresponding numbers of citations; “–” = not applicable.

4.1 Publication and Citation Structure of Urban Development and Management

Table 1 summarizes publication output and citation patterns across the study period. UDM research remained limited during much of the twentieth century. Only 83 documents were published between 1925 and 1975, followed by 2,827 during 1976–2000. Growth accelerated substantially after 2000. Annual output increased from 252 documents in 2001 to 1,147 in 2013, when it first exceeded 1,000 publications, and reached 2,096 in 2019. By 2025, annual output had risen to 5,562 documents, approximately 22 times the level recorded in 2001. This expansion was particularly pronounced in the most recent period: 18,550 documents were published between 2021 and 2025, accounting for 45.9% of the entire 1925–2025 dataset. Output increased by 31.6% between 2024 and 2025 alone.

The 40,435 documents had accumulated 1,285,702 citations by the date of data retrieval, corresponding to an overall average of 31.8 citations per document. Aggregate citations to documents published in a given year generally increased across successive publication cohorts until 2021, when papers published that year had accumulated 93,876 citations. Lower cumulative totals were observed for the subsequent cohorts, falling to 33,309 citations for documents published in 2025. This pattern should not be interpreted as evidence of declining research influence because recent publications have had considerably less time to accumulate citations. The same age effect is evident when recent publication cohorts are compared with older ones. Highly cited papers were particularly concentrated in the 2010–2017 period: 91 of the 158 documents with at least 500 citations and 18 of the 32 documents with at least 1,000 citations were published during these eight years. Thirteen of the 32 papers exceeding 1,000 citations were published between 2014 and 2017.

The citation-threshold distribution confirms the strong concentration of citation impact within a relatively small proportion of the literature. Only 32 documents (0.08%) had received at least 1,000 citations, 158 (0.39%) had received at least 500, and 2,594 (6.42%) had reached 100 citations. At the broader end of the distribution, 22,691 documents (56.12%) had received at least 10 citations and 29,089 (71.94%) at least five. Overall, 37,093 documents (91.73%) had been cited at least once, leaving 3,342 (8.27%) uncited at the time of data retrieval. Many of these uncited documents belonged to recent publication cohorts. The distribution therefore combines a small group of exceptionally highly cited publications with a much larger body of moderately cited work, a pattern that also underscores the importance of considering publication age when comparing citation performance.

4.2 Influential Papers on Urban Development and Management

Table 2 presents the 30 most cited documents in the UDM dataset together with their average C/Y, calculated with 2026 as the reference year. These results address RQ3 by identifying publications that have accumulated substantial citation influence within the retrieved literature. The most cited document was “Plastic waste inputs from land into the ocean” (2015), published in *Science*, with 11,110 citations. It was followed by “Internet of Things for smart cities” (2014), with 5,677 citations, and “A review on circular economy: The expected transition to a balanced interplay of environmental and economic systems” (2016), with 5,480 citations. The top 30 documents were distributed across multidisciplinary, medical, environmental, engineering, planning, and urban-studies journals. Three appeared in *The Lancet*, while multiple papers were published in outlets including *Science*, *IEEE Internet of Things Journal*, *Landscape and Urban Planning*, and *Urban Studies*. Most of these highly cited publications were published between 2007 and 2017.

The subject matter of these documents illustrates the disciplinary breadth captured by the UDM dataset. One group addresses environmental pollution and resource pressures, including marine plastic waste, soil contamination, microplastics in urban rivers, water-quality degradation, and the role of urban forests in pollution mitigation. A second group links urban conditions with public health through health equity, climate change, extreme heat, and greenspace. A third concerns smart cities and digital urbanism, including the IoT, big data, artificial intelligence, and critical perspectives on smart-city development. Green and blue infrastructure constitute another identifiable strand, represented by urban greening, sustainable drainage, nature-based solutions, urban biodiversity, and resilience-oriented approaches. Governance, planning, urban metabolism, and research on the size and form of cities further demonstrate the institutional and systems-oriented dimensions of the literature.

Citation rates per year provide an additional perspective because cumulative citation totals strongly favor older publications. Apart from the 2015 plastic-waste study, the highest C/Y values were recorded for the circular-economy review (548.0), the IoT study (473.1), and the 2021 review of heat extremes and health in *The Lancet* (382.4). These were followed by the study linking greenspace and health (255.0) and the 2021 review of water-quality degradation (243.4). The appearance of two papers published in 2021 among the 30 most cited documents, despite their comparatively short citation windows, indicates rapid citation accumulation around the heat–health and water-quality dimensions of contemporary urban research.

The composition of Table 2 also reveals an important limitation of the broad UDM search domain. Several highly cited documents, including studies concerning insect pathogens, lithium recycling, and mangrove forests, lie near the conceptual boundary of UDM rather than representing its central urban-management literature. Their presence reflects the breadth of the retrieved corpus and should therefore be considered when interpreting document-level

rankings. For this reason, the most cited documents are treated as indicators of the citation environment represented in the dataset rather than as a definitive canon of UDM scholarship.

Table 2. The 30 most cited documents in UDM literature (Scopus)

R	Title	Source	Year	TC	C/Y
1	Plastic waste inputs from land into the ocean	<i>Science</i>	2015	11,110	1,010.0
2	Internet of Things for smart cities	<i>IEEE Internet of Things Journal</i>	2014	5,677	473.1
3	A review on circular economy: The expected transition to a balanced interplay of environmental and economic systems	<i>Journal of Cleaner Production</i>	2016	5,480	548.0
4	Closing the gap in a generation: Health equity through action on the social determinants of health	<i>The Lancet</i>	2008	3,279	182.2
5	Urban greening to cool towns and cities: A systematic review of the empirical evidence	<i>Landscape and Urban Planning</i>	2010	2,655	165.9
6	Managing the health effects of climate change: Lancet and University College London Institute for Global Health Commission	<i>The Lancet</i>	2009	2,432	143.1
7	Exploring pathways linking greenspace to health: Theoretical and methodological guidance	<i>Environmental Research</i>	2017	2,295	255.0
8	Smart cities of the future	<i>European Physical Journal: Special Topics</i>	2012	2,137	152.6
9	Hot weather and heat extremes: Health risks	<i>The Lancet</i>	2021	1,912	382.4
10	Present state and future of the world's mangrove forests	<i>Environmental Conservation</i>	2002	1,701	70.9
11	Recovery and recycling of lithium: A review	<i>Separation and Purification Technology</i>	2017	1,666	185.1
12	Soil quality: A concept, definition, and framework for evaluation (A guest editorial)	<i>Soil Science Society of America Journal</i>	1997	1,642	56.6
13	SUDS, LID, BMPs, WSUD and more—The evolution and application of terminology surrounding urban drainage	<i>Urban Water Journal</i>	2015	1,636	148.7
14	Governance innovation and the citizen: The Janus face of governance-beyond-the-state	<i>Urban Studies</i>	2005	1,524	72.6
15	Contamination features and health risk of soil heavy metals in China	<i>Science of the Total Environment</i>	2015	1,477	134.3
16	Insect pathogens as biological control agents: Back to the future	<i>Journal of Invertebrate Pathology</i>	2015	1,345	122.3
17	Microplastic is an abundant and distinct microbial habitat in an urban river	<i>Environmental Science & Technology</i>	2014	1,344	112.0
18	Nature-based solutions to climate change mitigation and adaptation in urban areas: Perspectives on indicators, knowledge gaps, barriers, and opportunities for action	<i>Ecology and Society</i>	2016	1,270	127.0
19	Why India cannot plan its cities: Informality, insurgence and the idiom of urbanization	<i>Planning Theory</i>	2009	1,262	74.2
20	SWAT2000: Current capabilities and research opportunities in applied watershed modelling	<i>Hydrological Processes</i>	2005	1,234	58.8
21	The size, scale, and shape of cities	<i>Science</i>	2008	1,233	68.5
22	Various natural and anthropogenic factors responsible for water quality degradation: A review	<i>Water (Switzerland)</i>	2021	1,217	243.4
23	An information framework for creating a smart city through Internet of Things	<i>IEEE Internet of Things Journal</i>	2014	1,204	100.3
24	Smartmentality: The smart city as disciplinary strategy	<i>Urban Studies</i>	2014	1,191	99.3
25	Biodiversity in the city: Key challenges for urban green space management	<i>Frontiers in Ecology and the Environment</i>	2017	1,185	131.7

26	Temporal and spatial changes in social vulnerability to natural hazards	<i>Proceedings of the National Academy of Sciences of the United States of America</i>	2008	1,169	64.9
27	From fail-safe to safe-to-fail: Sustainability and resilience in the new urban world	<i>Landscape and Urban Planning</i>	2011	1,109	73.9
28	The changing metabolism of cities	<i>Journal of Industrial Ecology</i>	2007	1,060	55.8
29	On big data, artificial intelligence and smart cities	<i>Cities</i>	2019	1,041	148.7
30	Urban forests and pollution mitigation: Analyzing ecosystem services and disservices	<i>Environmental Pollution</i>	2011	1,039	69.3

Note: UDM = urban development and management; R = rank; TC = total citations; C/Y = citations per year; IEEE = Institute of Electrical and Electronics Engineers; SUDS = sustainable urban drainage systems; LID = low-impact development; BMPs = best management practices; WSUD = water-sensitive urban design; SWAT2000 = Soil and Water Assessment Tool 2000.

Table 3. The 30 most productive authors in UDM research

R	Author Name	Institution	Country	TP	TC	H	C/P	D1	D2	D3
1	C. Y. Jim	The Education University of Hong Kong	Hong Kong, China	47	3,223	28	68.57	15	15	17
2	L. Salvati	Sapienza University of Rome	Italy	40	2,403	21	60.08	0	9	31
3	M. Wang	Guangzhou University	China	36	1,530	20	42.5	0	0	36
4	A. Deletic	University of New South Wales	Australia	32	2,117	25	66.16	0	18	14
5	D. Butler	University of Exeter	United Kingdom	31	4,249	21	137.06	6	12	13
6	Z. Yang	Guangdong University of Technology	China	28	1,669	21	59.61	0	13	15
7	H. Jia	Tsinghua University	China	28	1,653	21	59.04	0	6	22
8	W. Rauch	University of Innsbruck	Austria	27	1,439	20	53.3	1	15	11
9	T. Yigitcanlar	Queensland University of Technology	Australia	26	3,133	23	120.5	0	11	15
10	R. R. Brown	Monash University	Australia	26	1,987	18	76.42	1	18	7
11	B. Chen	Beijing Normal University	China	26	1,315	18	50.58	0	13	13
12	D. Haase	Helmholtz Centre for Environmental Research-UFZ	Germany	25	3,050	19	122	0	4	21
13	C. Zevenbergen	IHE Delft Institute for Water Education	Netherlands	25	1,336	15	53.44	0	10	15
14	G. Liu	Beijing Normal University	China	25	1,063	16	42.52	0	7	18
15	S. Pauleit	Technical University of Munich	Germany	24	3,558	16	148.25	1	1	22
16	C. Fang	University of Chinese Academy of Sciences	China	23	2,546	19	110.7	0	3	20
17	F. Li	Tsinghua University	China	23	1,784	17	77.57	2	11	10
18	P. Zhao	Peking University	China	21	1,511	17	71.95	0	13	8
19	L. Shen	Hangzhou City University	China	21	1,332	16	63.43	0	4	17
20	M. Scholz	Independent Researcher	Germany	21	941	12	44.81	2	10	9
21	A. Rajabifard	University of Melbourne	Australia	21	656	14	31.24	0	4	17
22	H. Wang	Nanjing University	China	21	374	12	17.81	0	0	21
23	T. D. Fletcher	University of Melbourne	Australia	20	4,562	19	228.1	0	13	7
24	A. Sharifi	Hiroshima University	Japan	20	2,748	16	137.4	0	1	19
25	P. M. Bach	Monash University	Australia	20	1,337	17	66.85	0	6	14
26	Y. Zhang	Beijing Normal University	China	20	567	14	28.35	0	9	11
27	X. Zhang	University of Hong Kong	Hong Kong, China	19	2,111	17	111.11	0	6	13
28	N. B. Chang	University of Central Florida	United States	19	1,292	15	68	6	4	9
29	Y. Cai	Guangdong University of Technology	China	19	957	14	50.37	0	3	16
30	P. B. Cobbinah	University of Melbourne	Australia	19	827	14	43.53	0	1	18

Note: UDM = urban development and management; R = rank; TP = total publications; TC = total citations; H = h-index; C/P = citations per publication; D1 = publications during 1925–2005; D2 = publications during 2006–2015; D3 = publications during 2016–2025.

4.3 Leading Authors, Institutions, and Countries

Table 3 presents the 30 most productive authors in the UDM dataset according to TP. Additional indicators include TC, h-index, citations per publication, and publication output across three periods. C. Y. Jim ranked first with 47 publications and 3,223 citations, followed by L. Salvati with 40 publications, M. Wang with 36, A. Deletic with 32, and D. Butler with 31. Authors affiliated with Chinese institutions formed the largest national group in the table (12), followed by Australia (7), Germany (3), and Hong Kong (2), with additional representation from Italy, the United Kingdom, Austria, the Netherlands, Japan, and the United States. Several institutions—including Tsinghua University, Beijing Normal University, Guangdong University of Technology, Monash University, and the University of Melbourne—were represented by more than one highly productive author, indicating identifiable institutional concentrations of UDM research.

Productivity and citation influence did not follow the same ordering. T. D. Fletcher, for example, published 20 documents but accumulated 4,562 citations, corresponding to the highest citations-per-publication value in Table 3 (228.1). High C/P values were also recorded for S. Pauleit (148.25), A. Sharifi (137.4), D. Butler (137.06), D. Haase (122.0), and T. Yigitcanlar (120.5). Several of these authors work in areas that are prominent elsewhere in the dataset, including urban water management, green infrastructure, and smart or sustainable cities. By contrast, some highly productive authors had lower citation averages, including H. Wang (17.81 citations per publication) and Y. Zhang (28.35). These differences should be interpreted cautiously because publication timing varies substantially among authors and recent papers have had shorter citation windows.

The temporal distribution also shows a change in the composition of highly productive authors. C. Y. Jim was the only author among the top 30 with substantial output in all three periods, publishing 15 documents in 1925–2005, 15 in 2006–2015, and 17 in 2016–2025. Several Australian and European researchers, including A. Deletic, R. R. Brown, W. Rauch, and T. D. Fletcher, recorded substantial output during the middle period, particularly in research associated with urban water and related environmental systems. In contrast, many of the Chinese authors in Table 3 concentrated most or all of their publications in 2016–2025. This shift corresponds with the rapid overall expansion of UDM publications shown in Table 1 and indicates the growing contribution of Chinese research institutions during the most recent decade.

Table 4 extends the analysis to institutions. The University of the Chinese Academy of Sciences recorded the largest publication output, with 493 documents, followed by Beijing Normal University (424), Wuhan University (334), Tsinghua University (325), and the University of Melbourne (300), the highest-ranked institution outside China. Chinese institutions accounted for 13 of the 30 entries, in addition to two institutions from Hong Kong. Australia was represented by the University of Melbourne, Monash University, University of New South Wales, the University of Queensland, and the University of Sydney. Other highly represented national research systems included the United States, the Netherlands, Canada, the United Kingdom, Singapore, and Iran. The institutional distribution is consistent with the author-level results, particularly the strong representation of Beijing Normal University, Tsinghua University, the University of Melbourne, Monash University, and University of New South Wales.

Institutional productivity again differed from citation influence. Beijing Normal University accumulated the highest TC in Table 4 (26,845) and the highest h-index (75), whereas the University of the Chinese Academy of Sciences, despite ranking first in publication output, recorded 22,521 citations and an h-index of 74. University College London had the highest citation average, at 80.0 citations per publication, followed by the University of California, Berkeley (73.44), Arizona State University (69.32), the University of Sydney (65.74), the University of Melbourne (63.71), and Beijing Normal University (63.31). Lower citation averages were observed for several institutions whose publication portfolios were concentrated more heavily in the recent period, including Southeast University (23.41), Hohai University (30.21), the China University of Geosciences (31.41), and Tongji University (32.27).

The temporal distribution reveals a substantial geographical shift in institutional participation. During 1925–2005, the largest outputs among the institutions listed in Table 4 were recorded by the University of Hong Kong (39 publications), the University of Toronto (29), University College London (25), the University of Florida (19), and the University of California, Berkeley (18). During 2006–2015, Monash University (81 publications), the University of Melbourne (70), and Beijing Normal University (67) were among the leading contributors. The pattern changed markedly in 2016–2025, when the University of the Chinese Academy of Sciences produced 452 documents, Beijing Normal University 354, and Wuhan University 308. More than 90% of the listed output of several Chinese institutions—including Wuhan University, Southeast University, Hohai University, and the China University of Geosciences—fell within this latest period. By comparison, institutions such as the University of California, Berkeley and the University of Hong Kong showed a more distributed publication record across the three periods.

Country-level results are presented in Table 5. China ranked first in publication output with 9,309 documents, representing approximately 23% of the dataset, followed by the United States (7,850), the United Kingdom (3,318), Australia (2,392), and India (2,027). Canada (1,670), Italy (1,593), Germany (1,429), Spain (1,314), and the Netherlands (1,167) completed the top ten. Although the United States produced approximately 1,459 fewer

documents than China, it accumulated more TC (348,371 versus 297,595) and recorded a higher h-index (223 versus 190). Its citation average was also higher, at 44.38 citations per publication compared with 31.97 for China. The United Kingdom and Australia recorded h-indices of 172 and 147, respectively, and citation averages exceeding 52 citations per publication.

Table 4. The 30 most productive and influential institutions in UDM

R	Institution	Country	TP	TC	H	C/P	D1	D2	D3
1	University of Chinese Academy of Sciences	China	493	22,521	74	45.68	2	39	452
2	Beijing Normal University	China	424	26,845	75	63.31	3	67	354
3	Wuhan University	China	334	12,459	58	37.3	2	24	308
4	Tsinghua University	China	325	14,598	66	44.92	4	43	278
5	University of Melbourne	Australia	300	19,113	65	63.71	12	70	218
6	Tongji University	China	282	9,101	52	32.27	0	24	258
7	Peking University	China	279	14,757	64	52.89	6	60	213
8	University College London	United Kingdom	259	20,719	59	80	25	50	184
9	Delft University of Technology	Netherlands	250	12,493	59	49.97	14	60	176
10	Zhejiang University	China	242	8,578	52	35.45	3	25	214
11	University of Hong Kong	Hong Kong, China	237	11,864	60	50.06	39	39	159
12	Nanjing University	China	234	11,040	59	47.18	3	33	198
13	The Hong Kong Polytechnic University	Hong Kong, China	220	12,537	60	56.99	10	27	183
14	Monash University	Australia	217	13,440	54	61.94	11	81	125
15	University of New South Wales	Australia	203	9,802	47	48.29	17	37	149
16	Southeast University	China	199	4,659	37	23.41	1	10	188
17	National University of Singapore	Singapore	197	10,491	56	53.25	17	22	158
18	University of Toronto	Canada	196	9,478	44	48.36	29	53	114
19	University of Tehran	Iran	194	6,815	43	35.13	4	26	164
20	China University of Geosciences	China	193	6,062	41	31.41	0	10	183
21	The University of Queensland	Australia	192	8,569	52	44.63	14	55	123
22	Sun Yat-sen University	China	187	7,185	46	38.42	1	12	174
23	University of British Columbia	Canada	184	11,272	48	61.26	15	55	114
24	Hohai University	China	184	5,558	40	30.21	0	10	174
25	Arizona State University	United States	177	12,269	54	69.32	9	51	117
26	Chongqing University	China	177	7,793	46	44.03	0	22	155
27	University of Florida	United States	175	7,279	37	41.59	19	40	116
28	Wageningen University & Research	Netherlands	174	10,169	52	58.44	12	37	125
29	University of California, Berkeley	United States	161	11,824	49	73.44	18	57	86
30	University of Sydney	Australia	159	10,452	44	65.74	8	45	106

Note: UDM = urban development and management; R = rank; TP = total publications; TC = total citations; H = h-index; C/P = citations per publication; D1 = publications during 1925–2005; D2 = publications during 2006–2015; D3 = publications during 2016–2025.

Citation intensity varied considerably among countries. Sweden recorded the highest C/P value among the 30 countries (72.09), followed by Switzerland (63.87), Australia (53.86), Belgium (53.31), Hong Kong (53.03), the United Kingdom (52.81), and the Netherlands (52.40). Lower averages were observed for several countries whose UDM publication output was more heavily concentrated in recent years, including India (27.28), Iran (27.43), Malaysia (27.28), Saudi Arabia (25.37), Turkey (24.60), Poland (19.23), Thailand (19.22), and Indonesia (14.96). These differences should not be interpreted as direct measures of research quality because national publication portfolios differ considerably in age, disciplinary composition, and citation opportunity.

The temporal indicators show a pronounced shift in the geography of UDM research. Of China's 9,309 publications, 8,395 were published during 2016–2025. The United States, by contrast, recorded 1,369 publications during 1925–2005, substantially more than any other country in the early period, followed by the United Kingdom (520) and Canada (215). The data therefore indicate an early concentration of UDM scholarship in established English-speaking research systems, followed by rapid growth in Chinese publication output during the most recent period.

Population-normalized indicators provide a different view of research intensity. Hong Kong recorded 88.22 publications per million inhabitants, followed by Australia (87.85), the Netherlands (63.26), Sweden (61.12), Singapore (59.77), Switzerland (54.95), and Portugal (52.43). In citations per million inhabitants, Australia ranked highest at 4,731.6, followed by Hong Kong (4,678.4), Sweden (4,406.1), and Switzerland (3,509.9). By comparison, India

and Indonesia recorded 1.37 and 1.74 publications per million inhabitants, respectively. Population normalization therefore highlights substantial differences between absolute publication volume and the intensity of UDM research relative to national population size.

Taken together, the country-level results reveal two distinct dimensions of the geographical structure of UDM research. Large research systems, particularly China and the United States, dominate absolute publication and citation volumes, whereas several smaller and highly urbanized economies record much greater research intensity after population normalization. At the same time, the relatively low population-normalized output of rapidly urbanizing countries in South Asia and the limited representation of African countries among the leading contributors point to a geographical imbalance between where much of the world's future urban growth is expected to occur and where UDM research is currently concentrated. This imbalance provides an important context for the country-level coupling patterns examined in Section 5 and for the research priorities concerning the Global South discussed later in the study.

Table 5. The most productive and influential countries in UDM

R	Country	TP	TC	H	C/P	D1	D2	D3	P/Po	C/Po
1	China	9,309	297,595	190	31.84	87	827	8,395	6.59	209.77
2	United States	7,850	348,371	223	44.38	1,369	2,314	4,167	22.49	998.10
3	United Kingdom	3,318	175,230	172	52.81	520	857	1,941	47.45	2505.7
4	Australia	2,392	128,828	147	53.86	215	732	1,445	87.85	4731.6
5	India	2,027	55,294	107	27.28	86	291	1,650	1.37	37.45
6	Canada	1,670	77,384	119	46.34	215	470	985	41.62	1928.5
7	Italy	1,593	76,556	116	48.06	34	310	1,249	27.03	1299.2
8	Germany	1,429	67,567	115	47.28	65	285	1,079	17.08	807.79
9	Spain	1,314	54,214	100	41.26	27	251	1,036	27.46	1133
10	Netherlands	1,167	61,149	116	52.40	93	294	780	63.26	3314.5
11	Brazil	969	25,819	74	26.64	31	175	763	4.54	120.90
12	Iran	801	21,970	69	27.43	7	124	670	8.60	235.81
13	Japan	800	28,390	79	35.49	69	167	564	6.53	231.89
14	France	796	35,323	89	44.38	56	206	534	11.93	529.22
15	South Africa	775	27,208	77	35.11	71	162	542	11.84	415.69
16	Poland	755	14,519	55	19.23	12	63	680	19.95	383.66
17	South Korea	748	25,742	78	34.41	21	130	597	14.5	498.88
18	Sweden	654	47,150	98	72.09	49	160	445	61.12	4406.1
19	Hong Kong, China	651	34,522	93	53.03	55	100	496	88.22	4678.4
20	Malaysia	637	17,379	63	27.28	10	100	527	17.51	477.64
21	Portugal	545	19,698	73	36.14	9	92	444	52.43	1895
22	Indonesia	502	7,511	42	14.96	14	32	456	1.74	26.09
23	Switzerland	495	31,617	85	63.87	37	128	330	54.95	3509.9
24	Saudi Arabia	466	11,822	56	25.37	19	24	423	13.25	336.18
25	Turkey	464	11,413	56	24.60	24	101	339	5.28	129.8
26	Greece	404	16,647	65	41.21	18	117	269	40.82	1682
27	Thailand	362	6,958	43	19.22	28	42	292	5.06	97.23
28	Mexico	359	13,556	60	37.76	24	76	259	2.7	101.93
29	Belgium	354	18,870	68	53.31	25	83	246	30.06	1602.6
30	Singapore	353	17,221	72	48.78	24	48	281	59.77	2915.9

Note: UDM = urban development and management; R = rank; TP = total publications; TC = total citations; H = h-index; C/P = citations per publication; D1 = publications during 1925–2005; D2 = publications during 2006–2015; D3 = publications during 2016–2025; P/Po = publications per million inhabitants; C/Po = citations per million inhabitants. Population estimates: Worldometer (2026).

5 Science-Mapping and Thematic Analysis of Urban Development and Management Research

Performance indicators describe the scale and citation profile of UDM research, but they provide only a limited view of how the field is intellectually and conceptually organized. Science mapping complements the metric analysis by examining relationships among journals, countries, and research topics within the bibliographic dataset. In distance-based network maps, the proximity of two items represents the strength of their association, while clusters identify groups of items that are more strongly connected to one another than to the remainder of the network. Such maps are particularly useful for a large and multidisciplinary field such as UDM, where relationships among more than 40,000 documents cannot be adequately represented through publication and citation rankings alone [8, 11, 12].

The upper-right portion of the network contains a substantial social-science and spatial-planning component. An orange cluster comprising *Cities*, *Urban Studies*, *Habitat International*, *Land Use Policy*, and *Journal of Transport Geography* represents urban studies, planning, land policy, and transport geography. A neighboring cyan cluster containing *Remote Sensing*, *Applied Geography*, and *Computers, Environment and Urban Systems* reflects the geospatial methods used to observe urban expansion and land-use change. Ecological and landscape research is concentrated toward the lower-right portion of the network, where *Landscape and Urban Planning* is associated with *Landscape Ecology*, *Biological Conservation*, and *Ecological Applications*, as well as multidisciplinary outlets such as *Nature*, *Science*, *Proceedings of the National Academy of Sciences (PNAS)*, and *PLoS ONE*. A related cluster centered on *Ecological Indicators* and *Ecosystem Services* represents ecosystem assessment and environmental valuation. Climate- and hazard-oriented journals, including *Global Environmental Change*, *Climatic Change*, *Natural Hazards*, and *International Journal of Disaster Risk Reduction*, form another recognizable component. Overall, the co-citation structure portrays UDM as a multidisciplinary field whose intellectual foundations are strongly anchored in environmental and water sciences while remaining connected to planning, geography, ecology, energy, climate, and hazard research.

Figure 3 presents the bibliographic coupling network of countries publishing UDM research. The network was generated in VOSviewer using thresholds of at least five documents and 100 citations per country. Two countries are bibliographically coupled when publications associated with them cite common references; stronger links therefore indicate greater similarity in the literature on which their research draws (RQ5). Node size represents the relative prominence of each country in the network, while link thickness indicates coupling strength. China and the United States form the two largest nodes and are connected by the strongest visible link, indicating substantial overlap in the references used by the world's two largest producers of UDM research. A second group of strongly connected countries includes the United Kingdom, Australia, India, Germany, Canada, and the Netherlands.

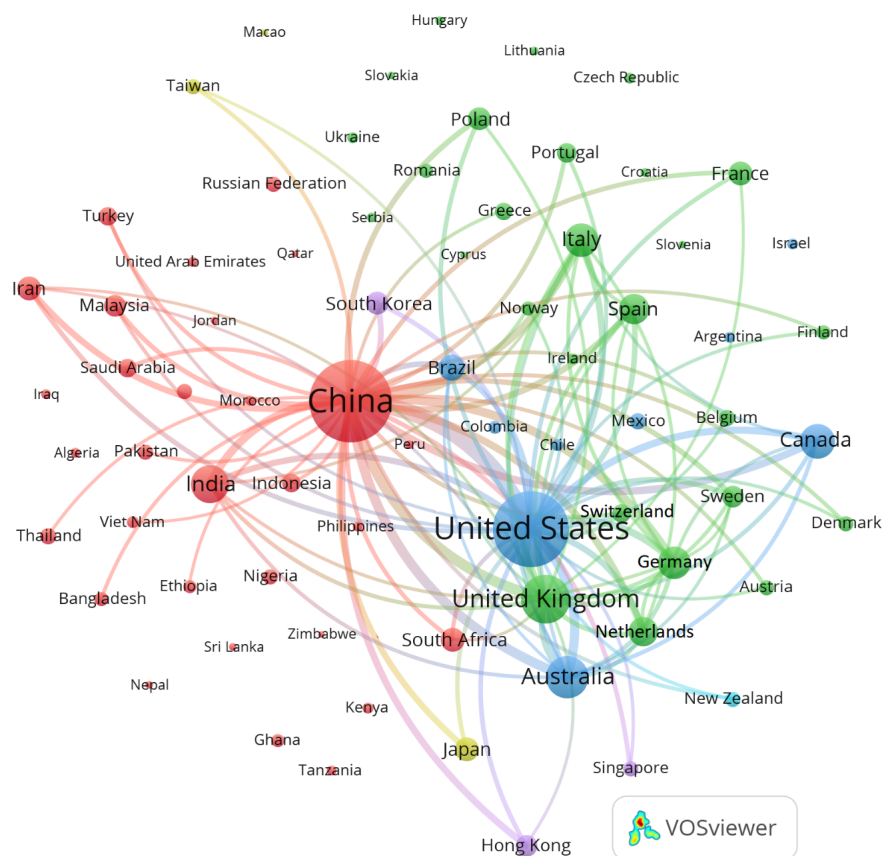


Figure 3. Bibliographic coupling network of countries publishing UDM research (minimum thresholds: five documents and 100 citations per country)

Note: UDM = urban development and management.

Approximately six clusters are visible, although these should be interpreted as bibliographic communities rather than rigid geopolitical divisions. The largest cluster contains India, Iran, Malaysia, Pakistan, Indonesia, Saudi Arabia, and Turkey, together with a wider group of countries in Asia, the Middle East, and Africa, including Bangladesh, Viet Nam, Nigeria, Ethiopia, Ghana, Kenya, and South Africa. Many of these countries are strongly connected with China

in the coupling network, indicating overlap in the references used across research concerned with rapidly changing urban systems, land conversion, and environmental pressures. This pattern does not establish a directional flow of knowledge from one country to another, but it does show that these national research literatures share substantial parts of their bibliographic foundations.

A European cluster includes the United Kingdom, Germany, the Netherlands, Italy, Spain, Sweden, and Switzerland, together with several Central and Eastern European countries, including Poland, Romania, the Czech Republic, Hungary, and countries in the Baltic and Balkan regions. Their relatively dense internal connections indicate substantial similarity in the literature cited across European UDM research. Another cluster contains the United States, Canada, and Australia together with Brazil, Mexico, Colombia, Chile, and Argentina, linking North American and Australian research with a substantial part of the Latin American literature represented in the network. Smaller clusters include Japan and a group containing highly urbanized East and Southeast Asian economies such as Taiwan, Hong Kong, and Singapore, while New Zealand occupies a separate but spatially proximate position to Australia.

At the broadest level, the network is organized around large and highly connected research systems associated with China, the United States, and Europe, while many countries in South Asia, Africa, and other regions occupy smaller or more peripheral positions. Here, “core” and “periphery” refer strictly to positions within the bibliographic coupling network rather than to the quality or importance of the research produced in individual countries. When considered alongside the population-normalized results in Section 4.3, the pattern indicates a geographical imbalance between the current concentration of indexed UDM scholarship and regions expected to experience a substantial share of future urban growth.

5.2 Keyword Structure and Thematic Development of Urban Development and Management

Figure 4 presents the keyword co-occurrence network generated in VOSviewer to examine the conceptual structure of UDM research (RQ6). Keywords are connected when they occur within the same publications; node size represents occurrence frequency, link strength reflects the frequency with which terms occur together, and colors distinguish clusters of closely associated terms. The most prominent and interconnected terms include “sustainable development,” “sustainability,” “urbanization,” “urban planning,” and “climate change.” Eight clusters can be distinguished, revealing both established areas of UDM research and groups of topics that have become increasingly prominent within the recent literature.

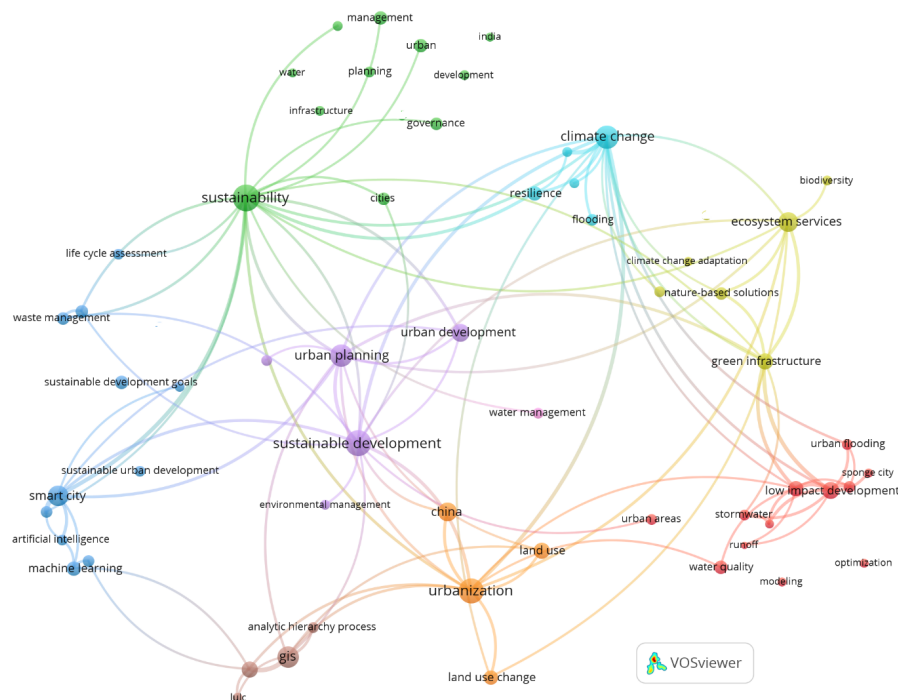


Figure 4. Co-occurrence network of keywords in UDM research (minimum thresholds: 200 occurrences and 100 links)

Note: UDM = urban development and management.

The purple cluster forms a broad conceptual center linking “sustainable development,” “urban planning,” “urban development,” “environmental management,” and “water management.” Its connections with a green cluster containing “sustainability,” “governance,” “infrastructure,” “planning,” and “cities” illustrate the close relationship between

environmental objectives and the policy and management dimensions of urban development. An orange cluster centered on “urbanization,” “China,” “land use,” and “land use change” reflects the substantial literature concerned with the spatial expansion and transformation of cities. This cluster is closely associated with a method-oriented group containing geographic information system (GIS), land use and land cover (LULC), and the analytic hierarchy process, indicating the prominent role of geospatial and multi-criteria methods in research on urban land-use change.

Other clusters capture more specialized and increasingly visible areas of research. One group connects “climate change” with “resilience” and “flooding” and is closely associated with nature-oriented terms including “ecosystem services,” “green infrastructure,” “nature-based solutions,” “climate change adaptation,” and “biodiversity.” Urban stormwater research forms a related group in which “low impact development” and “sponge city” occur alongside “urban flooding,” “stormwater,” “runoff,” “water quality,” “modeling,” and “optimization.” Their proximity suggests an increasingly integrated research stream concerned with climate adaptation and blue–green urban infrastructure. The presence of “sponge city” also reflects the importance of policy-driven approaches to urban water management within the Chinese research context.

A further cluster combines “life cycle assessment,” “waste management,” “sustainable urban development,” and “sustainable development goals” with terms such as “smart city,” “artificial intelligence,” and “machine learning.” Rather than demonstrating the emergence of these topics solely from network position, their prominence in the recent publication period, considered together with their co-occurrence relationships, identifies digital and data-supported urban management as an increasingly visible component of UDM research. This interpretation is consistent with the methodological definition adopted in Section 3, where emerging research directions are inferred from the combination of recent temporal prominence and science-mapping structure rather than from a separate topic-modeling or citation-burst analysis.

Table 6. Most frequent indexed keywords in UDM research (Scopus)

R	Keyword	TP	TC	H	C/P	D1	D2	D3
1	Sustainable development	7,539	293,393	205	38.92	361	1,444	5,734
2	Urban area	6,731	293,150	199	43.55	456	1,612	4,663
3	Urban planning	5,806	228,020	193	39.27	885	1,391	3,530
4	Urban development	5,142	189,594	170	36.87	227	1,318	3,597
5	Urbanization	3,899	179,858	182	46.13	250	808	2,841
6	Water management	3,635	151,087	166	41.56	234	981	2,420
7	Sustainability	3,630	154,298	163	42.51	100	800	2,730
8	Urban growth	3,255	135,121	154	41.51	76	605	2,574
9	Climate change	3,026	135,866	151	44.9	24	482	2,520
10	Land use	2,831	127,634	149	45.08	224	564	2,043
11	Decision making	2,540	96,484	128	37.99	132	559	1,849
12	Environmental management	2,304	101,536	144	44.07	217	531	1,556
13	Water supply	2,044	82,516	124	40.37	215	642	1,187
14	Water quality	1,867	83,254	123	44.59	188	515	1,164
15	Risk assessment	1,835	69,651	111	37.96	76	404	1,355
16	Cities	1,832	88,911	134	48.53	69	357	1,406
17	Environmental protection	1,789	82,993	128	46.39	261	397	1,131
18	Waste management	1,622	85,076	124	52.45	104	368	1,150
19	Urban population	1,601	55,022	105	34.37	373	465	763
20	Economic development	1,460	66,620	111	45.63	202	263	995
21	GIS	1,437	49,091	104	34.16	129	383	925
22	Runoff	1,387	62,417	115	45	88	328	971
23	Environmental monitoring	1,380	59,905	110	43.41	73	325	982
24	Land use change	1,370	55,252	113	40.33	34	290	1,046
25	Spatiotemporal analysis	1,314	37,998	88	28.92	4	95	1,215
26	Remote sensing	1,297	45,408	106	35.01	55	165	1,077
27	Environmental impact	1,273	66,980	125	52.62	206	287	780
28	Ecosystems	1,245	64,155	116	51.53	64	217	964
29	Stormwater	1,138	54,939	112	48.28	67	335	736
30	Ecosystem service	1,114	53,237	107	47.79	3	137	974

Note: UDM = urban development and management; R = rank; TP = total publications; TC = total citations; H = h-index; C/P = citations per publication; D1 = publications during 1925–2005; D2 = publications during 2006–2015; D3 = publications during 2016–2025; GIS = geographic information system.

Table 6 presents the 30 most frequent indexed keywords in the UDM dataset. “Sustainable development” was the most frequent term, occurring in 7,539 documents (18.6% of the dataset), which together accumulated 293,393 citations and an h-index of 205. It was followed by “urban area” (6,731 documents), “urban planning” (5,806), “urban development” (5,142), and “urbanization” (3,899). The remainder of the ten most frequent terms comprised “water management” (3,635), “sustainability” (3,630), “urban growth” (3,255), “climate change” (3,026), and “land use” (2,831). Together, these terms show that the conceptual center of UDM research lies at the intersection of urban growth, planning, sustainability, land management, and environmental pressures.

Several additional thematic groups can be distinguished among the 30 most frequent keywords. Water-related research is represented by “water management,” “water supply” (2,044 documents), “water quality” (1,867), “runoff” (1,387), and “stormwater” (1,138). Environmental management is represented by terms including “environmental management,” “environmental protection,” “waste management,” “environmental monitoring,” “environmental impact,” “ecosystems,” and “ecosystem service.” A third group captures spatial approaches to urban change through “land use change,” “spatiotemporal analysis,” “GIS,” and “remote sensing,” while “decision making,” “risk assessment,” and “economic development” represent management, policy, and socioeconomic dimensions.

Citation averages also vary across these keywords. Among the 30 terms in Table 6, the highest citations-per-publication values were associated with “environmental impact” (52.62), “waste management” (52.45), “ecosystems” (51.53), “cities” (48.53), and “stormwater” (48.28). Lower averages were recorded for several method-oriented terms, including “spatiotemporal analysis” (28.92), “GIS” (34.16), and “remote sensing” (35.01), as well as “urban population” (34.37). These differences describe citation patterns associated with documents containing particular indexed terms; they should not be interpreted as direct measures of the scientific importance of the corresponding research areas.

The temporal distribution provides further evidence of changing thematic emphasis. Terms with a comparatively substantial presence in the earliest period included “urban population” (373 documents), “urban planning” (885), “economic development” (202), “urbanization” (250), “urban area” (456), and “environmental protection” (261). By contrast, several themes became much more heavily concentrated in 2016–2025. For example, 2,520 of the 3,026 documents associated with “climate change,” 1,046 of 1,370 associated with “land use change,” 1,215 of 1,314 associated with “spatiotemporal analysis,” and 974 of 1,114 associated with “ecosystem service” were published during the most recent period. These distributions indicate a shift from an earlier emphasis on population, physical planning, and economic development toward climate, ecosystem, environmental-management, and spatial-analysis concerns. Because publication output increased sharply over time, these raw temporal counts are interpreted as evidence of changing prominence rather than as standalone growth rates.

The thematic map in Figure 5 was generated in bibliometrix using the 250 most frequent keywords, a minimum cluster frequency of five, and the Spinglass clustering algorithm [61]. The strategic diagram positions themes according to centrality and density. Centrality, represented on the horizontal axis, indicates the degree to which a thematic cluster is connected with other themes in the field, whereas density, represented on the vertical axis, reflects the internal cohesion of the cluster. Node size represents the relative frequency of the terms contained within each thematic group. The resulting map contains four principal clusters distributed across the four quadrants.

The largest cluster, organized around “sustainable development,” “urban area,” and “urban planning,” occupies the basic-themes quadrant. Its relatively high centrality and lower density indicate that these concepts are strongly connected with the wider UDM literature but encompass broad and internally diverse lines of research. The motor-themes quadrant contains a water-oriented cluster centered on “water management,” “water supply,” and “water quality.” Its combination of relatively high centrality and high density indicates both strong internal development and extensive connections with other parts of the UDM knowledge structure. This position is consistent with the high frequency of water-related terms in Table 6 and their visibility elsewhere in the science-mapping results.

The niche-themes quadrant contains a cluster centered on “United States,” “organization and management,” and “economics.” Its comparatively high density but lower centrality indicates an internally cohesive research area with more limited connections to the broader conceptual structure of UDM. In the emerging-or-declining quadrant, “environmental protection,” “waste management,” and “cities” form a cluster characterized by low centrality and low density. The strategic diagram alone cannot determine whether this cluster is emerging or declining; therefore, its position should be interpreted in conjunction with the temporal distributions in Table 6 rather than as independent evidence of thematic emergence.

Figure 6 provides a complementary descriptive view of the conceptual vocabulary of UDM. Word size represents keyword frequency, with “sustainable development” appearing most prominently, followed by “urban area,” “urban planning,” “urban development,” and “urbanization.” Their prominence confirms the central role of sustainability and spatial planning in the literature. A second group of frequently occurring terms—including “sustainability,” “urban growth,” “land use,” “climate change,” “decision making,” “water management,” and “environmental management”—shows that UDM extends beyond the physical expansion of cities to encompass environmental management, resource allocation, institutional decision making, and adaptation.

management, climate adaptation, spatial analysis, and governance-oriented decision making. More recent terms such as “adaptive management,” “governance approach,” and “ecosystem service” also point to increasingly differentiated approaches to managing complex urban systems. These patterns provide the basis for the thematic synthesis in Section 6, where changes in the field are interpreted in relation to the broader development of UDM and the identification of recent research directions.

6 Main Findings and Research Agenda

This section integrates the performance-analysis results in Section 4 with the science-mapping results in Section 5 for the 40,435 Scopus-indexed journal articles and reviews published between 1925 and 2025. The findings are first synthesized in relation to the six research questions and then examined across the main thematic and geographical patterns identified in the dataset. The section concludes by outlining substantive priorities for future UDM research and methodological directions for subsequent bibliometric studies.

6.1 Synthesis of Findings by Research Question

RQ1—How has the publication and citation structure of UDM developed?

UDM has a long publication history but has expanded most rapidly in recent decades. Only 83 documents were published between 1925 and 1975 and 2,827 between 1976 and 2000. Annual output subsequently increased from 252 documents in 2001 to 5,562 in 2025, approximately 22 times the 2001 level. The 2021–2025 period alone accounted for 18,550 publications, or 45.9% of the entire dataset, while annual output increased by 31.6% between 2024 and 2025. These patterns show that the literature represented in the UDM dataset has entered a period of particularly rapid expansion.

RQ2—What are the principal citation characteristics of the field?

The 40,435 documents had accumulated 1,285,702 citations by the retrieval date, corresponding to an average of 31.8 citations per document. Citation impact was highly concentrated: 91.73% of documents had been cited at least once, but only 158 (0.39%) had received at least 500 citations and 32 (0.08%) had exceeded 1,000 citations. Publications from 2010–2017 were particularly prominent among these highly cited works, accounting for 91 of the 158 documents with at least 500 citations and 18 of the 32 with at least 1,000 citations. Lower cumulative citation totals among the most recent publication cohorts largely reflect their shorter citation windows and should not be interpreted as evidence of declining influence.

RQ3—Which documents have accumulated the greatest citation influence?

The 30 most cited documents span a broad range of disciplinary settings, including multidisciplinary science, medicine and public health, engineering, environmental science, planning, and urban studies. Their subject matter can be organized into five broad groups: (i) environmental pollution and resource flows, including ocean plastics, soil contamination, microplastics, water-quality degradation, and the circular economy; (ii) urban environment and public health, including health equity, climate change, heat extremes, and greenspace; (iii) smart cities and digital urbanism, including the Internet of Things, big data, and artificial intelligence; (iv) green and blue infrastructure and resilience, including urban greening, sustainable drainage, nature-based solutions, and safe-to-fail approaches; and (v) governance, planning, urban metabolism, and urban form. The disciplinary breadth of these documents reflects the wide knowledge base represented in the UDM dataset, although several highly cited papers lie near the conceptual boundary of the search domain and should therefore be interpreted cautiously.

RQ4—How are leading authors, institutions, and countries distributed?

Research productivity was widely distributed at the author level. C.Y. Jim was the most productive author in the dataset with 47 publications, while the top 30 included 12 authors affiliated with Chinese institutions and seven with Australian institutions. Productivity and citation influence did not necessarily coincide: several authors with moderate publication counts, including T.D. Fletcher, S. Pauleit, A. Sharifi, D. Butler, D. Haase, and T. Yigitcanlar, recorded high citation averages. At the country level, China produced the largest number of documents (9,309; approximately 23% of the dataset), whereas the United States accumulated the largest number of citations (348,371) and the highest h-index (223). Population-normalized indicators produced a different pattern, with several smaller or highly urbanized economies—including Hong Kong, Australia, the Netherlands, Sweden, Singapore, and Switzerland—showing comparatively high research intensity. Sweden recorded the highest citations per publication among the 30 countries, followed by Switzerland. These results demonstrate that absolute research volume, citation impact, and population-normalized research intensity represent distinct dimensions of UDM research activity.

RQ5—What does science mapping reveal about the intellectual and geographical structure of UDM?

Journal co-citation analysis indicates that UDM draws on a multidisciplinary intellectual foundation. Environmental and water sciences occupy prominent positions through journals such as *Science of the Total Environment*, *Journal of Cleaner Production*, and *Journal of Environmental Management*. These sources are connected with identifiable research foundations in planning and urban studies, geospatial science, landscape ecology and ecosystem services, energy, climate, and hazards. Country bibliographic coupling reveals another dimension of this structure. China, the

United States, and several European research systems occupy large and highly connected positions in the network, while many countries in South Asia, Africa, and other regions occupy smaller or more peripheral positions. These patterns represent similarities in cited knowledge bases and should not be interpreted as direct evidence of one-way knowledge dependence.

RQ6—How is the conceptual structure of UDM organized, and which research directions have become increasingly prominent?

“Sustainable development” was the most frequent indexed keyword, occurring in 7,539 documents (18.6% of the dataset), followed by “urban area,” “urban planning,” “urban development,” and “urbanization.” The thematic map positioned the cluster centered on sustainable development and urban planning within the basic-themes quadrant, characterized by relatively high centrality and lower density. Water management occupied the motor-themes quadrant, with comparatively high centrality and density, while the cluster involving economics, organization and management, and the United States showed the high-density/low-centrality profile associated with niche themes. The cluster containing environmental protection, waste management, and cities occupied the low-centrality/low-density emerging-or-declining quadrant; its position alone does not distinguish emergence from decline. When the network structure is interpreted together with temporal keyword distributions, climate adaptation, blue–green infrastructure, nature-based solutions, and digitally supported urban management emerge as increasingly prominent recent directions. These include topics associated with sponge cities, low-impact development, artificial intelligence, machine learning, smart cities, life-cycle assessment, and the SDGs.

6.2 Thematic Development of Urban Development and Management Research

The combined evidence from keyword frequencies, temporal distributions, co-occurrence networks, and thematic mapping indicates a broad change in the thematic composition of UDM research. Earlier work was more strongly associated with urban population, physical planning, and economic development, whereas recent research increasingly connects urban development with environmental sustainability, climate adaptation, ecosystem management, spatial analysis, and digital technologies. This does not imply that earlier concerns have disappeared. Rather, established planning and development questions have become embedded within a broader set of environmental, technological, and governance challenges.

Sustainable development and urban planning remain among the most prominent concepts in the field. Their high centrality and comparatively low density in the thematic map indicate that they function as broad organizing themes connected with many other areas of UDM rather than as narrowly bounded research specialties. Water management presents a different pattern. The combination of “water management,” “water supply,” “water quality,” “runoff,” and “stormwater,” together with the prominence of hydrology and water-related journals in the co-citation network, places urban water research among the most internally developed and widely connected components of the UDM knowledge structure.

Spatial analysis represents another important line of development. Land-use change, GIS, remote sensing, and spatiotemporal analysis became much more prominent during 2016–2025, reflecting the increasing role of spatially explicit methods in observing urban expansion, environmental change, and land-management processes. This methodological development is particularly visible alongside the substantial recent contribution of Chinese research to the UDM literature.

Climate adaptation and blue–green infrastructure form a further area of increasing prominence. Keywords related to ecosystem services, nature-based solutions, resilience, sponge cities, low-impact development, flooding, and stormwater occur in closely connected parts of the keyword network. Their relationships suggest growing integration between urban water management, ecological infrastructure, and climate adaptation. Urban environment and health also gained visibility, particularly through research on heat extremes, greenspace, and water quality.

Digital approaches constitute another increasingly visible research stream. Smart cities, artificial intelligence, machine learning, and related data-driven approaches are connected with sustainability, infrastructure, and urban-management themes. Their growing presence suggests that digital technologies are increasingly being incorporated into established questions of planning, resource management, and environmental monitoring rather than developing as an entirely separate branch of UDM.

The temporal patterns therefore suggest three broad stages in the development of the literature represented in the dataset. The period up to 2005 was more strongly associated with population, planning, and economic-development concerns. During 2006–2015, water-sensitive urban development, green infrastructure, sustainability, and smart-city research became more visible, and many of the highly cited publications identified in Section 4 appeared during or around this period. From 2016 to 2025, climate change, ecosystem services, spatiotemporal analysis, the SDGs, nature-based solutions, and digital technologies became substantially more prominent. Taken together, these patterns indicate a gradual expansion of UDM from a predominantly growth- and planning-oriented literature toward one increasingly concerned with how urban development can be monitored, governed, and adapted within environmental and climatic constraints.

6.3 Cross-Cutting Patterns and Geographical Imbalances

Several cross-cutting patterns emerge when the performance and science-mapping results are considered together. First, publication volume and citation influence are not equivalent. China recorded the largest publication output, whereas the United States accumulated the highest TC and h-index. Population-normalized indicators produced another ordering, with several smaller and highly urbanized economies recording particularly high publication or citation intensity. These differences show why national research performance cannot be represented adequately by a single bibliometric indicator.

Second, the geographical distribution of UDM research is uneven. South Asian and African countries are less prominent among the leading contributors than China, North America, Europe, and Australia. South Africa is the only African country among the 30 most productive countries in Table 5, while India and Indonesia recorded 1.37 and 1.74 publications per million inhabitants, respectively. Many countries in South Asia and Africa also occupy smaller or more peripheral positions in the bibliographic coupling network. These findings indicate under-representation within the Scopus-indexed UDM literature analyzed here; they do not, by themselves, establish differences in national research capacity or one-way dependence on particular knowledge-producing countries. Nevertheless, the pattern is important because many of these regions are expected to experience substantial future urban growth.

Third, the results show that research intensity can differ markedly from absolute publication volume. Hong Kong, Australia, the Netherlands, Sweden, Singapore, and Switzerland all recorded high population-normalized publication rates despite large differences in national population and total output. Such patterns suggest that country size alone does not explain the geographical distribution of UDM scholarship, although the present bibliometric analysis does not test the institutional, economic, or policy factors underlying these differences.

Fourth, urban water research occupies an unusually prominent position across several independent analyses. Water-related keywords are frequent, the thematic map identifies water management as a motor theme, water and hydrology journals form a recognizable component of the co-citation network, and several highly cited authors work in related areas. The convergence of these results provides stronger evidence for the centrality of urban water management than any individual indicator alone.

Fifth, several recent publications accumulated citations rapidly despite relatively short citation windows. The 2021 studies concerning heat extremes and health and water-quality degradation, for example, entered the group of highly cited documents within only a few years. Together with the recent keyword patterns, this suggests increasing scholarly attention to interactions among climate, health, water, and urban environmental management.

Finally, the co-citation network indicates that UDM remains multidisciplinary rather than fully integrated around a single disciplinary core. Environmental science, hydrology, planning, geography, ecology, energy, and climate-related journals form recognizable but interconnected components of the knowledge base. This structure is consistent with the breadth of the UDM concept itself and suggests that future advances may increasingly depend on stronger connections among research streams that have historically developed within different disciplinary traditions.

6.4 Research Agenda for Urban Development and Management

The findings point to several priorities for future UDM research. A first priority concerns the operationalization of urban sustainability. Sustainable development occupies a highly central but comparatively low-density position in the thematic map, indicating that it connects widely across UDM while remaining conceptually broad. Future studies can strengthen this area by linking sustainability goals to measurable city-level indicators, including indicators associated with SDG 11, and by examining how such measures perform across different urban contexts.

A second priority is stronger integration among water management, ecological systems, spatial planning, and climate adaptation. The prominence of water management and the increasing visibility of ecosystem services, nature-based solutions, sponge cities, and low-impact development indicate substantial scope for research that connects water-sensitive design with blue-green infrastructure and wider planning decisions. Such work can move beyond evaluating individual interventions toward examining how combinations of ecological and engineered systems function at neighborhood, city, and metropolitan scales.

A third direction concerns the relationship between climate adaptation and urban health. The citation performance of recent studies on heat extremes, greenspace, and water quality, together with the increasing prominence of climate-related keywords, points to a growing need to examine how climate risks interact with public health and urban environmental conditions. Comparative studies across different climatic, socioeconomic, and urban-development contexts would be particularly useful.

A fourth priority concerns digital urban management. Artificial intelligence, machine learning, smart-city technologies, GIS, remote sensing, and spatiotemporal analysis are increasingly visible within the recent literature. Future work can move beyond descriptive monitoring toward explanatory and decision-oriented applications, including multi-city comparisons and the integration of spatial observation with planning and management processes. The governance of AI-enabled urban systems—including transparency, accountability, data quality, and institutional

responsibility—also warrants closer examination as digital technologies become more deeply embedded in urban decision making.

A fifth priority is the geographical imbalance identified in the dataset. The relatively limited representation of many South Asian and African countries is particularly important given the scale of expected urban growth in these regions. Future UDM research can place greater emphasis on cities in the Global South, not simply by transferring frameworks developed elsewhere, but by examining locally specific forms of urbanization, infrastructure provision, informality, environmental risk, governance, and adaptation.

Finally, the connections among waste management, environmental protection, resource use, and sustainability suggest opportunities to examine urban circularity more explicitly. Rather than assuming that the low-centrality/low-density cluster containing environmental protection and waste management is necessarily declining, future research can investigate whether these concerns are being reorganized within newer frameworks such as the circular economy, resource efficiency, and integrated urban metabolism.

6.5 Directions for Future Bibliometric Research

The present analysis also identifies several methodological opportunities for future bibliometric studies of UDM. First, subsequent research could combine Scopus with Web of Science, Dimensions, or OpenAlex to assess the extent to which database coverage influences the observed intellectual and geographical structure of the field. Such comparisons would be particularly useful for evaluating the representation of regional journals and research from countries that appear less prominently in the present dataset.

Second, future studies could apply explicitly temporal methods—including trend-topic analysis, thematic-evolution analysis, and topic-modeling approaches such as BERTopic or latent Dirichlet allocation—to examine how specific research themes emerge, transform, converge, or decline over time. These methods would provide a stronger temporal basis for distinguishing genuinely emerging themes from established themes that simply occupy low-centrality or low-density positions in a static thematic map.

Third, field- and time-normalized citation indicators could complement raw citation counts and citations per year. Measures such as field-weighted citation impact would permit more balanced comparisons between older and newer publications and among research streams with different citation practices.

Fourth, more focused bibliometric studies could refine search strategies around the recent directions identified here, including sponge cities, nature-based solutions, urban heat and health, blue-green infrastructure, circular urban systems, and AI-enabled urban management. Such studies could use narrower domain definitions and more detailed content-based methods to examine mechanisms and research questions that cannot be resolved through a field-wide bibliometric analysis.

7 Conclusion

This study provides a century-scale bibliometric and science-mapping assessment of UDM research based on 40,435 Scopus-indexed journal articles and reviews published between 1925 and 2025. Performance analysis, journal co-citation analysis, country bibliographic coupling, keyword co-occurrence analysis, and thematic mapping were combined to examine the growth, citation structure, leading contributors, intellectual foundations, conceptual organization, and changing thematic emphasis of the field.

The results show that UDM research expanded markedly over the study period, with 45.9% of the documents in the dataset published between 2021 and 2025. Citation influence was considerably more concentrated than publication output, with a relatively small group of publications accounting for the highest citation levels. Geographically, China recorded the largest publication output, whereas the United States accumulated the highest TC and h-index. Population-normalized indicators revealed a different pattern, with several smaller and highly urbanized economies showing comparatively high levels of research intensity.

The science-mapping results demonstrate the multidisciplinary character of UDM. Environmental and water sciences occupy prominent positions in its intellectual structure, while planning, geography, ecology, energy, climate, and hazard research form distinct but interconnected components. Sustainable development and urban planning function as broad foundational themes, whereas water management represents a comparatively well-developed and highly connected thematic area. The combined temporal and science-mapping evidence further shows the increasing prominence of climate adaptation, blue-green infrastructure, nature-based solutions, ecosystem-oriented approaches, spatial analysis, and digitally supported urban management.

Across the century covered by the dataset, the thematic emphasis of UDM has broadened substantially. Earlier research was more strongly associated with population growth, physical planning, and economic development, whereas recent work increasingly integrates these established concerns with environmental limits, climate resilience, ecosystem management, geospatial monitoring, and digital technologies. UDM therefore appears not as a single disciplinary field but as an increasingly interconnected research domain concerned with how urban development can be planned, monitored, governed, and adapted under changing environmental, technological, and social conditions.

7.1 Implications

7.1.1 Theoretical implications

The study provides an integrated representation of research streams that are often examined separately, including urban water management, land-use change, green and blue infrastructure, smart cities, environmental management, and urban governance. Their positions within the co-citation, keyword, and thematic structures show that UDM is organized around several interconnected but distinguishable knowledge domains rather than a single unified theoretical framework.

The thematic position of sustainable development is particularly informative. Its high centrality and comparatively low density indicate that sustainability connects widely across UDM research while encompassing internally diverse lines of inquiry. This pattern does not in itself demonstrate the absence of an integrating theory, but it suggests that sustainability functions primarily as a broad organizing concept whose operational meaning varies across planning, infrastructure, environmental management, governance, and technological applications. The research priorities identified in Section 6.4 therefore point toward opportunities for stronger conceptual integration, particularly around measurable sustainability outcomes, climate adaptation, blue–green infrastructure, urban health, circular urban systems, and digital governance.

The study also shows that thematic development in UDM involves increasing interaction among research traditions that historically developed with different disciplinary foundations. Water management, ecological infrastructure, climate adaptation, geospatial analysis, and digitally supported urban management increasingly intersect with established planning and governance questions. This convergence provides a basis for future research that treats cities as coupled environmental, infrastructural, technological, and institutional systems rather than as collections of separate sectoral problems.

7.1.2 Practical implications

The findings also have implications for urban planning and management practice. The prominence of water management, blue–green infrastructure, climate adaptation, nature-based solutions, and spatial analysis reflects sustained attention to managing environmental pressures within rapidly changing urban systems. For urban planners and managers, these research streams provide a broad evidence base for approaches including water-sensitive urban design, sponge-city strategies, nature-based solutions, ecosystem-service planning, and spatially informed environmental management.

The growing visibility of artificial intelligence, machine learning, smart-city technologies, remote sensing, and spatiotemporal analysis also indicates an expanding role for digital tools in urban monitoring and decision support. Their practical relevance, however, extends beyond technical capability. As these approaches become more closely integrated with urban decision making, questions concerning data quality, transparency, institutional responsibility, and the governance of digitally supported decisions become increasingly important.

The geographical results identify another practical concern. Many rapidly urbanizing countries in South Asia and Africa remain comparatively less represented in the Scopus-indexed literature analyzed here, while a substantial share of publication and citation activity is concentrated in China, North America, Europe, and Australia. This imbalance supports greater attention to locally grounded UDM research in rapidly urbanizing regions and to international collaboration that allows planning, infrastructure, environmental, and governance approaches to be examined across different urban contexts rather than assumed to transfer directly from one setting to another.

Finally, the maps of leading journals, research themes, countries, and intellectual connections provide researchers and practitioners with a structured overview of the UDM knowledge base. Rather than identifying where future researchers are “most likely to make an impact,” the analysis helps locate established research communities, increasingly prominent themes, and areas where stronger connections across disciplines and geographical contexts may be particularly valuable.

7.2 Limitations

Several limitations should be considered when interpreting the findings. First, the analysis relies exclusively on Scopus. The results therefore reflect the database’s journal coverage, indexing policies, document classification, and metadata structure. Publications indexed only in other databases, as well as books, conference papers, other excluded document types, and non-English publications, are not represented. The dataset should consequently be understood as a large Scopus-based representation of UDM research rather than a complete record of all scholarship in the field.

Second, the deliberately broad search strategy was designed to capture the multidisciplinary knowledge environment surrounding UDM, but this breadth also introduced documents located near the conceptual boundaries of the field. This is particularly visible among several highly cited publications concerning topics such as insect pathogens, lithium recycling, and mangrove forests. No post-retrieval manual exclusion was applied on the basis of perceived relevance to individual documents, thereby avoiding subjective document-level selection but retaining some peripheral literature within the corpus. Consequently, document-level rankings and some keyword patterns should be interpreted as

representing the wider bibliographic environment of UDM rather than a narrowly defined canon of urban-management scholarship. Future studies addressing specific UDM subdomains could employ more restrictive search strategies or predefined topical screening criteria.

Third, bibliographic databases are dynamic. The present results represent Scopus as accessed on 30 June 2026, while the publication window was deliberately restricted to complete publication years from 1925 through 2025. Citation counts, affiliations, indexing records, and other metadata may change after the retrieval date. Citation-based indicators are also strongly affected by publication age, disciplinary citation practices, and document type. Older publications have had more time to accumulate citations than recent papers, particularly those published after 2021. For this reason, cumulative citation counts should not be interpreted as direct measures of research quality.

Fourth, author and institutional names may be subject to disambiguation problems because institutional mergers, name variants, changes in affiliation, and similarly named authors can affect productivity rankings. Country-level counting may also assign the same internationally co-authored publication to multiple countries, depending on the counting procedure, and should therefore be interpreted as participation in research output rather than as mutually exclusive national contributions.

Fifth, the thematic analysis identifies relationships among keywords and their positions within the conceptual structure of the dataset, but static co-occurrence and thematic maps cannot by themselves establish whether a low-centrality or low-density theme is emerging or declining. The interpretation of recent research directions in this study therefore combines network position with temporal keyword distributions rather than relying on the thematic map alone. More explicit temporal approaches, including thematic-evolution analysis, trend-topic analysis, citation-burst detection, and topic modelling, could provide finer-grained evidence of thematic change.

Finally, bibliometric methods describe the structure, visibility, and development of a research literature but do not directly evaluate the methodological quality, theoretical validity, or practical effectiveness of the individual studies represented in the dataset. The patterns identified here should therefore be regarded as a map of the UDM research landscape rather than as an assessment of the scientific quality of particular publications, authors, institutions, countries, or research themes.

Author Contributions

Conceptualization, M.S. and J.M.M.; methodology, M.S. and J.M.M.; software, M.S.; validation, M.S. and J.M.M.; formal analysis, M.S.; investigation, M.S.; data curation, M.S.; writing—original draft preparation, M.S.; writing—review and editing, M.S. and J.M.M.; visualization, M.S.; supervision, J.M.M.; project administration, J.M.M. All authors have read and agreed to the published version of the manuscript.

Data Availability

The bibliographic data analyzed in this study were retrieved from the Scopus database. Details of the search strategy, retrieval date, screening criteria, and data selection procedure are provided in Section 3. The processed data supporting the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

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